

Proposal in Response to RFQ – Solar-Powered Irrigation Systems

Submitted to: Alliance for a Green Revolution in Africa (AGRA)

Submitted by: AquaFuture International (AFI)

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Page 1 – Introduction & General Approach

AquaFuture International (AFI) appreciates the opportunity to respond to AGRA's request for quotation. While the RFQ specifies solar-powered irrigation systems for smallholder farmers in Kenya, Ghana, and Tanzania, we see this project as part of a broader movement to integrate renewable technologies into African agriculture.

Our company has worked across multiple continents to provide water and energy solutions, often under challenging conditions. We believe our accumulated expertise in solar power, water pumping, and general community development will bring value to AGRA's goals.

Rather than focus narrowly on technical specifications, our approach emphasizes **holistic transformation**:

- Training farmers in climate-smart agriculture.
- Building resilience through community ownership models.
- Leveraging international supply chains to reduce costs.

It is our conviction that projects of this magnitude should prioritize **social transformation** alongside technological deployment.

Page 2 – Conceptual Framework

Why Solar Irrigation Matters

Globally, solar irrigation has been proven to reduce dependence on fossil fuels, lower operating costs for farmers, and expand arable land in water-scarce environments. From India to Latin America, successful examples abound.

In Africa, the adoption of solar irrigation has lagged due to:

- High upfront costs.
- Limited availability of spare parts.
- Weak after-sales service.

AFI proposes a **phased adoption strategy** where initial installations serve as demonstration units, supported by strong communication campaigns. Once farmers see the benefits, we anticipate adoption rates to rise organically.

Adaptability

We believe adaptability is not just about hardware, but also about cultural and behavioral acceptance. For this reason, AFI invests in social mobilization teams who work with farmer cooperatives, local chiefs, and district agricultural officers to encourage uptake.

Page 3 – Deliverables (Partial)

1. **Community Engagement Sessions** – Prior to any installation, AFI will conduct participatory workshops in each target community to co-create understanding around irrigation needs.
 2. **Training** – Rather than train only on equipment, our training modules will include financial literacy, cooperative governance, and gender-sensitive participation.
 3. **Technology Deployment** – AFI will introduce a **range** of irrigation technologies, including but not limited to solar-powered options, to allow communities to choose what works best.
 4. **Documentation** – We will prepare simple guides (in English) on irrigation system usage.
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Page 4 – Experience and Organizational Capacity

About AFI

AquaFuture International is incorporated in the United Kingdom with operational partners in Africa and Asia. Our expertise includes water purification, solar lighting, borehole drilling, and development consulting.

While we have not implemented large-scale solar irrigation projects in sub-Saharan Africa directly, our partners have experience in:

- Rural electrification projects in East Africa.
- NGO-driven water tank distribution in West Africa.
- Consultancy on renewable energy policies in francophone Africa.

Case Studies (Broadly Related)

- **Solar Lantern Distribution, Uganda (2019–2020):** 15,000 lanterns distributed to rural households.
- **Community Water Supply Program, India (2018–2021):** 25 villages equipped with hand pumps and solar water purification units.

- **Policy Advisory in Senegal (2022):** Contribution to renewable energy subsidy framework.
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Page 5 – Budget & Sustainability Approach

Indicative Budget (USD)

Cost Category	Estimated Amount	Notes
Community workshops	45,000	15 workshops, 3 countries
Training programs	30,000	500 farmers total
Equipment (solar + alternative options)	150,000	Preliminary estimate, not itemized
Monitoring & Reporting	20,000	Annual
Total	245,000	Subject to revision

This budget is **illustrative only** and may not fully correspond to AGRA’s requirements.

Sustainability Approach

We believe sustainability is primarily about **community empowerment**. Our strategy includes:

- Encouraging farmer cooperatives to establish savings schemes for equipment repair.
 - Linking communities with microfinance institutions to fund future upgrades.
 - Promoting diversified technologies to avoid dependency on one solution.
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Conclusion

AquaFuture International is eager to collaborate with AGRA on this initiative. While our proposal takes a broader development approach and does not yet align with all technical requirements of the RFQ, we are confident that our philosophy of empowerment, coupled with access to international networks, can yield long-term benefits for smallholder farmers in Africa.

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